

Conforming Bernstein–Bézier Quasi-Trefftz Spaces for Variable-Coefficient Elliptic Problems

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Abstract

We introduce a conformity-first Bernstein–Bézier quasi-Trefftz construction for variable-coefficient PDEs. The method starts from the globally assembled continuous polynomial spline space and only then imposes equation-dependent residual moments, so continuity is exact by construction. The Bernstein domain-point representation makes the reduction explicit in arbitrary spatial dimension, with residual moments determining selected B-coefficients while the remaining coefficients form a reduced equation-adapted space. For second-order elliptic operators, testing the residual against polynomials of a fixed degree below the trial degree gives a closed-form local dimension, the difference of two simplex polynomial dimensions, which lowers the polynomial complexity of an element from volume order to trace order. That reduction is one of representation size; the prototype solves the constrained problem as a saddle point system, which is larger than the parent problem, so no solver speedup is claimed here. Under stable range-lifting and noncharacteristic assumptions, the reduced spaces retain the approximation order of the parent polynomial space. Two- and three-dimensional variable-anisotropic experiments illustrate the construction and confirm the dimension accounting against that closed form in every case computed. They also separate the two residual rules: testing three degrees below the trial degree tracks the accuracy of the parent space throughout, while testing two degrees below removes more coordinates but loses accuracy as the polynomial degree rises and as elements lose shape quality.

Keywords: Bernstein–Bézier polynomials, quasi-Trefftz methods, conforming finite elements, variable-coefficient PDEs, anisotropic diffusion, dimension reduction

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1. Introduction

Trefftz methods reduce the size of high-order approximation spaces by incorporating the governing differential equation into the discrete trial space. The idea goes back to Trefftz [25], and its modern form for wave problems begins with the ultra-weak variational formulation of Cessenat and Després [5], in which plane-wave solutions of the constant-coefficient Helmholtz operator serve as local basis functions. The state of that theory, including approximation and conditioning of plane-wave spaces, is surveyed by Hiptmair, Moiola and Perugia [10].

Classical Trefftz spaces require local exact solutions, which are generally unavailable once the coefficients vary. Two responses have developed. The first replaces the exact Trefftz property by an approximate one. Generalized plane waves satisfy the equation up to a prescribed order at a point [12], and polynomial quasi-Trefftz spaces impose vanishing Taylor coefficients of the residual, which has been carried out for the wave equation [14], the convected Helmholtz equation [15], and general smooth-coefficient elliptic operators [13]. The second response is algebraic: embedded

Trefftz methods take a full polynomial parent space, form the elementwise operator matrix, and use its kernel as the reduced space [20], so no analytic local solution is needed at all. A conformity-first variant of the same idea, in which exact C^1 Bernstein–Bézier smoothness relations are imposed before the equation-dependent reduction, has been developed for heterogeneous high-frequency Helmholtz problems [16]; the present paper treats the C^0 case in arbitrary dimension and makes the reduction counts explicit.

Both families are built on discontinuous parent spaces and recover interelement coupling through discontinuous Galerkin numerical fluxes. That is a deliberate choice, because a locally reduced space is in general not continuous across faces. It is also the point at which the present construction departs from them.

Equation-adapted local spaces are of course not confined to the Trefftz literature. Multiscale finite element methods build basis functions from local solves of the same operator [11], and localized orthogonal decomposition constructs a corrected coarse space whose basis decays exponentially away from its patch [21]. Those methods target coefficient heterogeneity at scales the mesh cannot resolve and pay for their basis with subgrid solves. The construction below targets a different quantity, the polynomial complexity of a high-degree element, and its local problems are finite-dimensional moment conditions rather than PDE solves.

The computational side rests on the Bernstein–Bézier basis. Beyond its role in spline theory [8, 19], the B-form supports optimal-complexity finite element assembly on simplices [2, 17], fast quadrature-based operator application [18], and pyramid algorithms in any dimension [1]. Two of its properties matter here: differentiation and degree reduction are sparse operations on coefficients, and continuity across a shared subsimplex is the identification of coefficients at shared domain points rather than a constraint to be enforced.

The present paper develops a conforming version of the Trefftz reduction in Bernstein–Bézier coordinates. The starting point is the globally continuous piecewise-polynomial space $S_p^0(\Delta)$ on a simplicial partition Δ . Shared Bernstein coefficients are identified first, using the standard domain-point description of Lai and Schumaker [19]. Equation-dependent residual moment conditions are imposed only after this C^0 assembly. Thus the method is conformity first and quasi-Trefftz second. This ordering is important: continuity never has to be repaired by a discontinuous Galerkin penalty or by an auxiliary trace variable, and the reduced space remains a subspace, or an affine subset in the nonhomogeneous case, of the usual conforming polynomial space.

The main contribution is the explicit construction and dimension accounting in arbitrary spatial dimension. If the polynomial degree is p and residual moments are tested against $\mathbb{P}_{p-m}$, the number of locally free coefficients for the second-order construction is

$$N_{d,p,m} = \binom{p+d}{d} - \binom{p-m+d}{d}.$$

For fixed d and m , this is $O(p^{d-1})$, compared with $O(p^d)$ coefficients in the full polynomial space. The reduction is therefore volume-to-trace in polynomial degree. For the choice $m = 3$ used in the numerical experiments, the local dimensions are $3p$ in two dimensions and $(3p^2 + 3p + 2)/2$ in three dimensions. We also derive the corresponding global formulas in the assembled C^0 space.

A second point is that the global residual system is treated rank-aware. The mathematical dimension is the dimension of the parent C^0 space minus the rank of the assembled residual map. No positive-definiteness assumption is made on the residual Gram matrix, and no normal equations are required. For the approximation theory we isolate the only genuinely global ingredient: a stable right inverse on the range of the assembled C^0 residual operator. Under this range-stability

property, a standard conforming interpolant can be corrected into the quasi-Trefftz space without losing approximation order.

The numerical study covers variable-coefficient problems in two and three dimensions. Structured and perturbed-Delaunay meshes confirm the exact rank formulas in both. The more aggressive $p - 2$ residual rule can overcompress on irregular meshes, whereas $p - 3$ closely tracks the accuracy of the full C^0 parent space while removing a substantial fraction of its global unknowns. A Fichera-corner computation demonstrates the construction on a nonconvex three-dimensional geometry with anisotropic diffusion, advection and reaction.

The paper is organized as follows. Section 2 states the model problem and recalls Bernstein–Bézier notation. Section 3 gives the local residual-moment construction and its reduction formulas in general dimension. Section 4 defines the assembled C^0 quasi-Trefftz space and derives exact global dimension formulas. Section 5 gives a compact approximation result under range stability. Section 6 compares the construction with quasi-Trefftz discontinuous Galerkin methods, embedded Trefftz methods, static condensation and multiscale bases. Section 7 discusses rank-aware construction and sum-factorized residual evaluation. Section 8 presents the numerical experiments.

2. Model problem and Bernstein–Bézier notation

Let $\Omega \subset \mathbb{R}^d$, $d \geq 2$, be a bounded polyhedral domain. We consider

$$\mathcal{L}u := -\nabla \cdot (A(x)\nabla u) + b(x) \cdot \nabla u + c(x)u = f \quad \text{in } \Omega, \quad (1)$$

with standard Dirichlet, Neumann or Robin boundary conditions. We assume throughout that

$$A \in W^{1,\infty}(\Omega)^{d \times d}, \quad b \in L^\infty(\Omega)^d, \quad c \in L^\infty(\Omega),$$

and that the symmetric part A_s is uniformly elliptic:

$$\xi^T A_s(x) \xi \geq a_0 |\xi|^2 \quad \text{for a.e. } x \in \Omega, \quad (2)$$

with $a_0 > 0$. These assumptions are sufficient for the residual moments used below. Additional conditions needed for coercivity or a Gårding inequality enter only when the Galerkin problem is considered.

Let Δ be a shape-regular simplicial partition of Ω . For a simplex

$$T = \langle v_0, \dots, v_d \rangle \in \Delta$$

with barycentric coordinates $\lambda_0, \dots, \lambda_d$, let

$$\alpha = (\alpha_0, \dots, \alpha_d) \in \mathbb{N}_0^{d+1}, \quad |\alpha| = p.$$

The degree- p domain point associated with α is

$$\xi_\alpha = \frac{1}{p} \sum_{i=0}^d \alpha_i v_i, \quad (3)$$

and the corresponding Bernstein polynomial is

$$B_\alpha^p = \frac{p!}{\alpha_0! \cdots \alpha_d!} \prod_{i=0}^d \lambda_i^{\alpha_i}. \quad (4)$$

Following [19], we write $\mathcal{D}_{p,T}$ for the degree- p domain points on T and index a B-coefficient by its domain point. Thus every $s \in \mathbb{P}_p(T)$ has a unique B-form

$$s|_T = \sum_{\xi \in \mathcal{D}_{p,T}} c_\xi B_\xi^T. \quad (5)$$

The domain points are geometric labels for the coefficients; in general $c_\xi \neq s(\xi)$.

The global C^0 spline space is

$$S_p^0(\Delta) = \{s \in C^0(\overline{\Omega}) : s|_T \in \mathbb{P}_p(T) \ \forall T \in \Delta\}. \quad (6)$$

The continuity condition is particularly simple in B-form: B-coefficients associated with the same domain point on a common subsimplex are identified. If Δ_k denotes the set of k -dimensional simplices in the mesh, then the number of degree- p domain points in the relative interior of a k -simplex is $\binom{p-1}{k}$, and hence

$$\dim S_p^0(\Delta) = \sum_{k=0}^d |\Delta_k| \binom{p-1}{k}. \quad (7)$$

Essential boundary conditions are imposed by removing the corresponding boundary domain points from this count.

3. Local quasi-Trefftz construction in arbitrary dimension

Fix integers $m \geq 2$ and $p \geq m$, and set

$$r = p - m \geq 0. \quad (8)$$

The parameter m measures how far the residual test degree lies below the trial degree. The condition $p \geq m$ is assumed throughout, and the numerical method developed below uses $m = 3$.

Definition 3.1 (Local residual-moment quasi-Trefftz space). For $T \in \Delta$ and $f \in L^2(T)$, define

$$\mathcal{Q}_{p,m}(T; f) = \{s \in \mathbb{P}_p(T) : (\mathcal{L}s - f, q)_T = 0 \ \forall q \in \mathbb{P}_{p-m}(T)\}. \quad (9)$$

For $f = 0$ this is a linear space; for general f it is either empty or an affine translate of $\mathcal{Q}_{p,m}(T; 0)$.

The definition is basis independent. Bernstein domain points become useful when the space is parameterized. Choose the facet

$$e = \langle v_1, \dots, v_d \rangle = \{\lambda_0 = 0\}.$$

For $j = 0, \dots, p$, define the j th row of domain points parallel to e by

$$\mathcal{D}_{p,T}^j(e) = \{\xi_\alpha \in \mathcal{D}_{p,T} : \alpha_0 = j\}. \quad (10)$$

A stars-and-bars count gives

$$|\mathcal{D}_{p,T}^j(e)| = \binom{p-j+d-1}{d-1}. \quad (11)$$

Let

$$\mathcal{D}_{p,T}^{<m}(e) = \bigcup_{j=0}^{m-1} \mathcal{D}_{p,T}^j(e), \quad \mathcal{D}_{p,T}^{\geq m}(e) = \mathcal{D}_{p,T} \setminus \mathcal{D}_{p,T}^{<m}(e).$$

The second set corresponds exactly to

$$\text{span}\{B_\xi^T : \xi \in \mathcal{D}_{p,T}^{\geq m}(e)\} = \lambda_0^m \mathbb{P}_{p-m}(T). \quad (12)$$

Consequently,

$$|\mathcal{D}_{p,T}^{\geq m}(e)| = \binom{p-m+d}{d} = \dim \mathbb{P}_{p-m}(T). \quad (13)$$

Thus the number of selected deep B-coefficients agrees exactly with the number of residual moment conditions.

Let the coefficient vector be partitioned as $\mathbf{c}_T = (\mathbf{c}_{T,S}, \mathbf{c}_{T,I})^T$, where S denotes the first m rows and I the deep rows. With a basis $\{q_\beta\}$ of $\mathbb{P}_{p-m}(T)$, define

$$(R_T)_{\beta,\xi} = (\mathcal{L}B_\xi^T, q_\beta)_T, \quad (f_T)_\beta = (f, q_\beta)_T.$$

Then the local constraints are

$$R_{T,S}\mathbf{c}_{T,S} + R_{T,I}\mathbf{c}_{T,I} = \mathbf{f}_T. \quad (14)$$

The hypothesis of the next proposition is that $R_{T,I}$ is nonsingular. That is not an extra assumption on the mesh or the degree. It follows from ellipticity once the element is small enough, and the reason is visible in the leading term of the residual on the deep space.

Proposition 3.2 (Noncharacteristic deep block). *Let $A \in W^{1,\infty}(T)^{d \times d}$ with symmetric part satisfying (2), and let $b, c \in L^\infty(T)$. Write $R_{T,I} = P_T + E_T$, where P_T collects the principal part,*

$$(P_T)_{\beta\gamma} = \left(-A : D^2(\lambda_0^m q_\gamma), q_\beta \right)_T, \quad (15)$$

and E_T collects the terms of $\mathcal{L}$ of order at most one. Then:

- (i) *the facet e is noncharacteristic for $\mathcal{L}$, since $a_e := \nabla \lambda_0 \cdot A \nabla \lambda_0 \geq a_0 |\nabla \lambda_0|^2 > 0$ on T by ellipticity, and the term of $D^2(\lambda_0^m q)$ carrying the lowest power of λ_0 is $m(m-1)\lambda_0^{m-2} q \nabla \lambda_0 \otimes \nabla \lambda_0$, whose contribution to P_T is the positive definite weighted mass matrix*

$$(W_e)_{\beta\gamma} = \left(m(m-1)a_e \lambda_0^{m-2} q_\gamma, q_\beta \right)_T; \quad (16)$$

- (ii) *after affine scaling to a shape-regular reference simplex, $\|E_T\| \leq C h_T \|P_T\|$, with C depending on shape regularity and on the $W^{1,\infty}$ bounds of the data, because the principal part carries h_T^{-2} while the remaining terms carry at most h_T^{-1} ;*

- (iii) *consequently, if P_T is nonsingular then $R_{T,I}$ is nonsingular whenever*

$$h_T \leq h_0 := \left(C \kappa(P_T) \right)^{-1}, \quad (17)$$

where κ denotes the spectral condition number.

Proof. Since λ_0 is affine,

$$D^2(\lambda_0^m q) = m(m-1)\lambda_0^{m-2} q \nabla \lambda_0 \otimes \nabla \lambda_0 + m\lambda_0^{m-1} (\nabla \lambda_0 \otimes \nabla q + \nabla q \otimes \nabla \lambda_0) + \lambda_0^m D^2 q,$$

which gives (i) on contracting the first term with A and using ellipticity, the weight in (16) being strictly positive on the interior of T . Part (ii) is the usual affine scaling count, and (iii) is a Neumann series for $R_{T,I} = P_T(I + P_T^{-1} E_T)$ using $\|P_T^{-1} E_T\| \leq \kappa(P_T) \|E_T\| / \|P_T\|$. $\square$

Remark 3.3 (What Proposition 3.2 does and does not give). Two limitations should be stated plainly. First, W_e is only one of the three groups of terms in P_T ; the cross term $m\lambda_0^{m-1}(\nabla \lambda_0 \otimes \nabla q + \dots)$ carries the same power of h_T and is not a small perturbation of W_e . Measured on the planar problem, $\|P_T + W_e\| / \|P_T\|$ is 0.98 at $p = 5, m = 3$ and 1.00 at $p = 8, m = 3$, so the positivity in (i) identifies the facet as noncharacteristic but does not by itself prove P_T nonsingular. Freezing A at a point and substituting $y = A^{-1/2}x$ turns P_T into the corresponding block for $-\Delta$ on a transformed simplex, so its invertibility is a property of that simplex alone; it held in every configuration computed for this paper, over degrees $4 \leq p \leq 8$, both residual rules, all facets, and anisotropy ratios up to 100, and Section 8.5 reports the conditioning.

Second, h_0 in (17) is not uniform in the degree. Empirically $\|E_T\| / \|P_T\|$ is proportional to h_T with a constant that is mildly decreasing in p , between 8.1×10^{-3} at $p = 5$ and 4.6×10^{-3} at $p = 8$, while $\kappa(P_T)$ grows by roughly one order per two degrees. The resulting threshold (17) falls from about 0.7 at $p = 5, m = 3$ to about 3×10^{-3} at $p = 8, m = 3$, and faster still for $m = 2$ and for strong anisotropy, where $\kappa(P_T)$ reaches 3×10^8 at $p = 8$ with anisotropy ratio 100. Proposition 3.2 is therefore an existence statement at fixed p , not a uniform-in- p stability result, and the p -robustness of the reduction is an open question that the conditioning measurements of Section 8.5 quantify rather than settle.

Remark 3.4 (Facet choice, and what happens if the block is singular). Under Proposition 3.2 the conclusion holds for every facet, so (19) is independent of the facet used to parameterize the space; only the conditioning of $R_{T,I}$ depends on that choice, and Section 8.5 measures the spread. Nothing in the definition (9) refers to a facet: the layers are a parameterization, not part of the space. If $R_{T,I}$ happens to be singular for one facet, another facet may be used, and if no facet gives a nonsingular block then the local dimension exceeds (19) and the global statement to use is the rank-aware formula (30) rather than (31). The implementation detects this from the rank-revealing factorization of Section 7.

Proposition 3.5 (Layer elimination). *If $R_{T,I}$ is nonsingular, which Proposition 3.2 guarantees for h_T small enough, then every choice of the retained coefficients $\mathbf{c}_{T,S}$ determines a unique element of $\mathcal{Q}_{p,m}(T; f)$, with*

$$\mathbf{c}_{T,I} = R_{T,I}^{-1}(\mathbf{f}_T - R_{T,S}\mathbf{c}_{T,S}). \quad (18)$$

In particular,

$$\dim \mathcal{Q}_{p,m}(T; 0) = \binom{p+d}{d} - \binom{p-m+d}{d}. \quad (19)$$

Proof. The residual equations (14) form a square system for the deep coefficients because of (13). Nonsingularity gives (18). The remaining coefficients are free, giving (19). $\square$

The proposition should be read as a simultaneous elimination statement. There is no one-to-one pairing between an individual residual equation and an individual domain point.

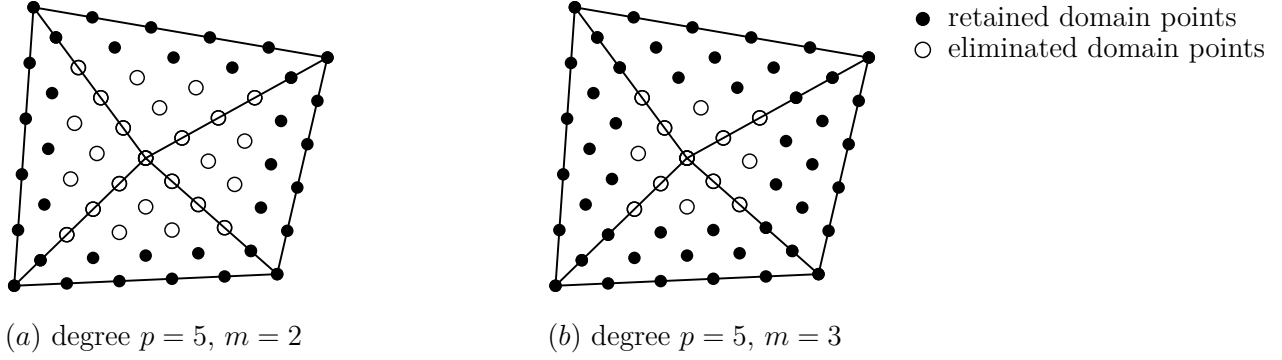


Figure 1: Bernstein parameterization of the local construction of Section 3, drawn on a four-triangle patch in $d = 2$ at degree $p = 5$. Panel (a) is the residual offset $m = 2$ and panel (b) is $m = 3$. On each triangle the rows $\mathcal{D}_{p,T}^j(e)$ of (10) with $j < m$, parallel to the selected outer edge, are retained, and the deeper rows $\mathcal{D}_{p,T}^{\geq m}(e)$ of (12) are determined by the residual moments. The counts of (19) give $\dim \mathbb{P}_5 = 21$ coordinates in the parent space, of which $21 - 10 = 11$ are retained in panel (a) and $21 - 6 = 15$ in panel (b). The quasi-Trefftz space is defined by the constraints (9); the picture only shows a convenient parameterization.

Corollary 3.6 (General-dimensional reduction formula). *Under the hypothesis of Proposition 3.5,*

$$N_{d,p,m} := \dim \mathcal{Q}_{p,m}(T; 0) = \binom{p+d}{d} - \binom{p-m+d}{d} \quad (20)$$

$$= \sum_{j=0}^{m-1} \binom{p-j+d-1}{d-1}. \quad (21)$$

For fixed d and m ,

$$N_{d,p,m} = \frac{m}{(d-1)!} p^{d-1} + O(p^{d-2}), \quad \frac{N_{d,p,m}}{\dim \mathbb{P}_p(T)} = \frac{md}{p} + O(p^{-2}). \quad (22)$$

For the choice $m = 3$,

$$N_{d,p,3} = \binom{p+d}{d} - \binom{p+d-3}{d}. \quad (23)$$

Two important specializations are

$$N_{2,p,3} = 3p, \quad N_{3,p,3} = \frac{3p^2 + 3p + 2}{2}. \quad (24)$$

Thus the local size drops from $O(p^2)$ to $O(p)$ in two dimensions and from $O(p^3)$ to $O(p^2)$ in three dimensions.

4. Global C^0 construction and reduction formulas

The global construction starts from $S_p^0(\Delta)$. No discontinuous local space is assembled first. For each element T , let

$$A_T : \mathbb{R}^{N_h} \rightarrow \mathbb{R}^{\dim \mathbb{P}_p(T)}$$

be the Boolean/restriction map from the vector C of assembled global B-coefficients to the local B-coefficients on T . Shared trace coefficients therefore appear only once in C .

Define

$$Y_h^{p-m} = \prod_{T \in \Delta} \mathbb{P}_{p-m}(T) \quad (25)$$

and the assembled residual map $R_h : S_p^0(\Delta) \rightarrow Y_h^{p-m}$ by

$$(R_h s, q_h) = \sum_{T \in \Delta} (\mathcal{L}s, q_T)_T. \quad (26)$$

In B-coordinates,

$$\mathbf{R}_h C = \begin{pmatrix} R_{T_1} A_{T_1} \\ \vdots \\ R_{T_{N_T}} A_{T_{N_T}} \end{pmatrix} C. \quad (27)$$

Definition 4.1 (Global conforming quasi-Trefftz space). The global affine space is

$$S_{p,m}^{0,qT}(\Delta; f) = \left\{ s \in S_p^0(\Delta) : (\mathcal{L}s - f, q)_T = 0 \ \forall q \in \mathbb{P}_{p-m}(T), \ \forall T \in \Delta \right\}. \quad (28)$$

This is C^0 by definition. If C_f is one solution of $\mathbf{R}_h C = F_h$ and Z_h is a basis of $\ker \mathbf{R}_h$, then

$$C = C_f + Z_h y \quad (29)$$

parameterizes the affine space. Hence the exact global dimension is

$$\dim S_{p,m}^{0,qT}(\Delta; f) = \dim S_p^0(\Delta) - \text{rank } \mathbf{R}_h, \quad (30)$$

whenever the affine equations are consistent. Formula (30), rather than a presumed full row rank, is the appropriate statement for a rank-aware implementation.

When the residual equations are independent, the rank is

$$\text{rank } \mathbf{R}_h = |\Delta_d| \binom{p-m+d}{d},$$

and (7) gives the following exact count.

Proposition 4.2 (Global full-row-rank formula). *If $\mathbf{R}_h$ has full row rank, then*

$$\dim S_{p,m}^{0,qT}(\Delta; f) = \sum_{k=0}^d |\Delta_k| \binom{p-1}{k} - |\Delta_d| \binom{p-m+d}{d}, \quad (31)$$

before essential boundary conditions.

For $m = 3$, (31) becomes

$$\dim S_{p,3}^{0,qT}(\Delta; f) = \sum_{k=0}^d |\Delta_k| \binom{p-1}{k} - |\Delta_d| \binom{p+d-3}{d}. \quad (32)$$

In two dimensions, the element-interior term cancels exactly:

$$\dim S_{p,3}^{0,qT}(\Delta; f) = |\Delta_0| + (p-1)|\Delta_1|. \quad (33)$$

In three dimensions, using $\binom{p}{3} = \binom{p-1}{3} + \binom{p-1}{2}$,

$$\dim S_{p,3}^{0,qT}(\Delta; f) = |\Delta_0| + (p-1)|\Delta_1| + \binom{p-1}{2} (|\Delta_2| - |\Delta_3|). \quad (34)$$

The latter identity is useful because it shows explicitly how the three-dimensional residual reduction removes all element-interior polynomial growth and an additional face-scale contribution.

5. Approximation under stable C^0 range lifting

The reduction formulas above are algebraic and do not require a global stability theorem. To state an approximation result without assuming full row rank, define

$$\mathcal{Y}_h = \text{range } R_h \subset Y_h^{p-m}$$

and let P_h be the L^2 -orthogonal projector onto $\mathcal{Y}_h$.

Assumption 5.1 (Stable range lifting). There exists a linear map $E_h : \mathcal{Y}_h \rightarrow S_{p,0}^0(\Delta)$ such that

$$R_h E_h q_h = q_h, \quad \|E_h q_h\|_{0,\Omega} + h |E_h q_h|_{1,\Omega} \leq C h^2 \|q_h\|_{0,\Omega}, \quad (35)$$

with C independent of h . Here $S_{p,0}^0(\Delta)$ denotes the subspace satisfying homogeneous essential boundary conditions.

The assumption concerns only the nonzero residual spectrum. In matrix terms it is a lower bound on the smallest nonzero singular value after the natural second-order scaling. This is compatible with rank-deficient residual matrices.

Theorem 5.2 (Approximation by the conforming quasi-Trefftz space). *Assume Assumption 5.1. Let $u \in H^s(\Omega)$, $2 \leq s \leq p+1$, solve (1) and satisfy the imposed essential boundary conditions. Suppose a conforming degree- p quasi-interpolant $I_h u \in S_p^0(\Delta)$ satisfies the usual local estimates through order two, for instance the Cl  ment or Scott–Zhang operator [4, 24]. Then there exists $Q_h u \in S_p^0(\Delta)$ such that*

$$R_h Q_h u = P_h \Pi_{p-m} f$$

and

$$|u - Q_h u|_{1,\Omega} \leq C h^{s-1} |u|_{s,\Omega}, \quad (36)$$

$$\|u - Q_h u\|_{0,\Omega} \leq C h^s |u|_{s,\Omega}. \quad (37)$$

If R_h has full row rank, $P_h = I$ and $Q_h u \in S_{p,m}^{0,qT}(\Delta; f)$.

Proof. Set

$$d_h = P_h (\Pi_{p-m} f - R_h I_h u).$$

Since $\mathcal{L}u = f$ and P_h is an orthogonal projection,

$$\|d_h\|_0 \leq \|\mathcal{L}(u - I_h u)\|_0 \leq C h^{s-2} |u|_s.$$

Define $Q_h u = I_h u + E_h d_h$. Then $R_h Q_h u = P_h \Pi_{p-m} f$ and (35) gives

$$|E_h d_h|_1 \leq C h^{s-1} |u|_s, \quad \|E_h d_h\|_0 \leq C h^s |u|_s.$$

Combining these estimates with the interpolation bounds proves (36)–(37). $\square$

For a coercive variational problem, the Galerkin method restricted to the affine space inherits the H^1 estimate from C  a’s lemma. Under the usual adjoint H^2 regularity, the standard duality argument yields the additional order in L^2 . For a G  rding-coercive operator the same conclusion follows, for sufficiently small h , by the standard Schatz argument; see, e.g., [4].

Remark 5.3 (The rank-deficient case). Theorem 5.2 produces $Q_h u$ with $R_h Q_h u = P_h \Pi_{p-m} f$, which lies in $S_{p,m}^{0,qT}(\Delta; f)$ only when $P_h = I$. If R_h is rank deficient the affine system $\mathbf{R}_h C = F_h$ is solvable if and only if $F_h \in \text{range } \mathbf{R}_h$, that is exactly when $P_h \Pi_{p-m} f = \Pi_{p-m} f$. When that fails the quasi-Trefftz set is empty and the natural substitute is the consistent problem $R_h s = P_h \Pi_{p-m} f$, whose solution set is a nonempty affine subspace of dimension (30); Theorem 5.2 then bounds the distance from u to that set, and the defect $(I - P_h) \Pi_{p-m} f$ is a computable consistency quantity. In every case computed for this paper $\mathbf{R}_h$ had full row rank, so the distinction did not arise; the rank and the tolerance used to determine it are reported in Section 8.5.

Remark 5.4 (Essential boundary conditions). Essential conditions are imposed by removing the corresponding boundary domain points from the coefficient vector. On an element with a face on the Dirichlet boundary this removes coefficients from the retained layers, never from the deep layers, because the deep set $\mathcal{D}_{p,T}^{\geq m}(e)$ of (12) consists of interior domain points once the facet e is chosen to be the boundary face. The elimination (18) is therefore unaffected and the moment constraints remain consistent with the prescribed trace. Choosing e to be a boundary face is always possible for a boundary element, and Remark 3.4 shows the dimension count does not depend on that choice. The experiments below use Robin conormal data, for which no coefficients are removed at all.

6. Relation to existing constructions

The reduction above sits between several established constructions, and the differences are worth stating precisely.

Quasi-Trefftz discontinuous Galerkin.. Polynomial quasi-Trefftz spaces [13–15] are defined elementwise by requiring the Taylor coefficients of $\mathcal{L}s - f$ at a point of the element to vanish up to a chosen order. That is a pointwise condition and it yields an explicit triangular recursion for a local basis. Definition 3.1 replaces the pointwise Taylor conditions by integrated moments against $\mathbb{P}_{p-m}(T)$. The two agree in counting, both remove $\binom{p-m+d}{d}$ local directions, but the moment form needs no expansion point, is invariant under affine change of coordinates, and is the natural object once the parent space has already been assembled. The essential structural difference is the parent space: the cited constructions reduce a discontinuous space and then reconnect elements through DG fluxes, whereas the present space is a subset of $S_p^0(\Delta)$ and is continuous by construction, so no penalty parameter, lifting operator or trace unknown appears.

Embedded Trefftz.. Embedded Trefftz discontinuous Galerkin methods [20] are the closest relative in spirit. They also start from a full polynomial space and obtain the reduced space algebraically, from the kernel of the elementwise operator matrix, rather than from analytic local solutions. The construction here shares that algebraic character and (30) is its rank-aware counterpart. It differs in where the reduction is applied. Embedded Trefftz reduces each element of a broken space and couples the results with DG terms; here the assembly is performed first and the residual map (26) acts on the assembled coefficients, so a coefficient shared by two elements is reduced once as a single global unknown rather than twice as two local ones.

Static condensation.. Eliminating element-interior degrees of freedom by static condensation is standard in high-order finite elements [23]. The distinction is that condensation is a solver transformation: the discrete space, and therefore the discrete solution, is unchanged, and the interior unknowns are recovered exactly from the Schur complement of the same bilinear form.

Here the space itself changes. The eliminated coefficients are determined by the strong residual of $\mathcal{L}$ rather than by the Galerkin system, the resulting subset of $S_p^0(\Delta)$ is smaller than the parent space, and the discrete solution differs from the parent Galerkin solution by an amount the experiments of Section 8 quantify. The layer picture of Figure 1 also differs from the interior-bubble picture of condensation: the retained coefficients are the first m rows parallel to a chosen facet, which include but are not limited to the trace.

Multiscale and equation-adapted bases.. Multiscale finite element methods [11] and localized orthogonal decomposition [21] also build spaces adapted to the operator, and the latter is likewise defined through a constraint on a residual. The purpose and the cost differ. Those methods address coefficient variation below the mesh scale and their basis functions solve local boundary value problems, with patch sizes chosen so that the localization error is controlled. The present reduction addresses the growth of the local polynomial dimension with the degree, its local problems are the square linear systems (18), and the coefficients are assumed resolved by the mesh. The two reductions are therefore complementary rather than competing, and nothing prevents applying the moment reduction inside a multiscale coarse space.

Higher regularity.. Nothing in the ordering requires C^0 . Bernstein–Bézier smoothness relations of order r are themselves linear conditions on coefficients at domain points near a shared subsimplex [19], so they can be imposed before the residual moments in exactly the same way. That is the route taken in [16] for heterogeneous high-frequency Helmholtz problems, where C^1 conformity is assembled first and the equation-dependent reduction follows. The counts change, because each additional order of smoothness removes coefficients that the residual moments would otherwise have been free to determine, and the dimension bookkeeping of Section 4 has to be redone against the smaller parent space.

What the construction buys.. Relative to a DG quasi-Trefftz method at the same degree, the conforming reduction removes the face terms and their parameters, and it removes the duplication of shared coefficients. Relative to the parent conforming method at the same degree, it removes rank $\mathbf{R}_h$ coordinates. The price is that the reduced space is defined through a constraint whose rank must be respected, which is why (30) rather than (31) is the primary statement, and that the discrete solution is no longer the Galerkin solution of the parent space. Section 8 measures that price.

7. Rank-aware implementation and residual evaluation

The construction separates three operations that should not be conflated.

First, C^0 assembly is performed in the Bernstein domain-point coordinates. The element-to-global map A_T identifies all shared vertex, edge and face coefficients before the residual constraints are formed. This is the standard B-form assembly used in high-order simplicial finite elements [2, 17], and the identification of coefficients at shared domain points is exactly the continuity description of [19].

Second, the residual matrix is compressed with rank-revealing linear algebra. If $\mathbf{R}_h$ is rank deficient, a QR factorization with column pivoting or an SVD determines its effective range [9]. The reduced basis is obtained from a null-space representation or from an equivalent constrained solve. No normal equations are required, and conditioning should be interpreted on the nonzero spectrum.

Table 1: Representative storage for dense tetrahedral Bernstein evaluation tables and the corresponding sum-factorized one-dimensional tables.

p	local DOFs	quadrature points	dense table (MB)	SF tables (MB)
10	286	2197	4.79	0.039
16	969	6859	50.71	0.133
20	1771	12167	164.40	0.243
24	2925	19683	439.25	0.402

Third, high-degree Bernstein evaluations can be carried out by collapsed-coordinate sum factorization, following the optimal-complexity Bernstein algorithms of [1, 2, 17, 18]. On a tetrahedron, with Duffy coordinates (t_1, t_2, t_3) [7], a Bernstein polynomial can be written as a nested product of one-dimensional Bernstein factors. Values, gradients and Hessians are then evaluated by successive one-dimensional contractions. The physical coefficients A , b , c and f may be evaluated directly at quadrature points; they need not be replaced by polynomial coefficient fields merely to use sum factorization.

For polynomial coefficient approximations and exact-enough quadrature, the residual matrix is integrated exactly. With nonpolynomial coefficients, quadrature gives a perturbation of the exact residual map. If the perturbation is smaller than the smallest nonzero scaled singular value, the rank and range-stability properties are preserved by standard singular-value perturbation estimates [9, Ch. 8].

Table 1 records the storage reduction of the verified tetrahedral sum-factorization kernel. The final column is the maximum difference from the dense Bernstein evaluation. The current reference implementation emphasizes verification rather than optimized timing, so only storage and agreement are reported here.

8. Numerical experiments

All meshes are shape regular in the sense of [4, 6], and the perturbed-Delaunay families below are used to test how the reduction responds when that constant degrades. The reduced problem is solved as a constrained minimization, so the linear system is of saddle point type and the solvers and preconditioners of [3] apply; the residual-space preconditioner reported with the supplementary code uses MINRES [22] on the indefinite system. The experiments are designed to test the construction and the dimension reduction rather than to compare unrelated discretization families. The full C^0 degree- p space is therefore used as the parent-space reference.

8.1. A smooth cube problem: h - and p -convergence

We first consider

$$-\Delta u + u = f \quad \text{in } (0, 1)^3$$

with the manufactured solution $u(x, y, z) = \exp(x + y + z)$ and consistent Robin boundary data. The cube is divided into six tetrahedra per Cartesian cell. The quasi-Trefftz rule uses $m = 3$, so that $r = p - 3$.

Table 2 shows a fixed- p refinement with $p = 4$. The qT approximation tracks the parent C^0 solution while using fewer global coordinates. Table 3 shows a degree sweep on the six-tetrahedron mesh; both spaces exhibit rapid convergence, while the dimension ratio decreases as p grows.

Table 2: Cube h -refinement with $p = 4$.

Cartesian n	tetrahedra	full DOFs	qT DOFs	full rel. L^2	qT rel. L^2
1	6	125	101	5.47×10^{-2}	6.73×10^{-2}
2	48	729	537	2.40×10^{-3}	2.95×10^{-3}
3	162	2197	1549	3.48×10^{-4}	4.37×10^{-4}

Table 3: Cube degree sweep on six tetrahedra.

p	full DOFs	qT DOFs	full rel. L^2	qT rel. L^2
4	125	101	5.47×10^{-2}	6.73×10^{-2}
6	343	223	5.70×10^{-3}	8.09×10^{-3}
8	729	393	3.53×10^{-4}	5.55×10^{-4}
10	1331	611	1.45×10^{-5}	2.43×10^{-5}
12	2197	877	4.22×10^{-7}	7.39×10^{-7}

8.2. Variable anisotropy and irregular tetrahedra

The next problem uses the full variable diffusion tensor

$$A(x, y, z) = \begin{pmatrix} 2 + 0.30x & 0.10 + 0.02z & 0.05 + 0.01y \\ 0.10 + 0.02z & 1.5 + 0.20y & 0.08 + 0.015x \\ 0.05 + 0.01y & 0.08 + 0.015x & 1 + 0.10z \end{pmatrix}$$

and reaction coefficient

$$c(x, y, z) = 1 + 0.15xy + 0.05z.$$

The exact solution remains $u = \exp(x + y + z)$ and the forcing and Robin data are generated analytically. We compare the residual degrees $p - 2$ and $p - 3$ on a 52-tetrahedron perturbed-Delaunay mesh.

At $p = 5$, the $p - 3$ space removes about 37% of the global C^0 coordinates while increasing the relative L^2 error by only about 20%. In contrast, the more aggressive $p - 2$ reduction removes about 74% but loses roughly one decimal digit of accuracy. This is the principal numerical reason for using $m = 3$ in the remainder of the paper.

Table 5 summarizes the observed dimensions for representative structured and perturbed-Delaunay meshes. In every case listed here, the measured quasi-Trefftz dimensions agree exactly with the full-row-rank prediction (31).

8.3. Two-dimensional variable anisotropy

The planar companion uses the same operator family on the unit square, with

$$A(x, y) = \begin{pmatrix} 2 + 0.30x & 0.10 + 0.02x \\ 0.10 + 0.02x & 1.5 + 0.20y \end{pmatrix}, \quad c(x, y) = 1 + 0.15xy, \quad (38)$$

exact solution $u = \exp(x + y)$, and Robin conormal data generated analytically. In two dimensions the local counts of (19) read

$$\dim \mathbb{P}_p - \dim \mathbb{P}_{p-2} = 2p + 1, \quad \dim \mathbb{P}_p - \dim \mathbb{P}_{p-3} = 3p,$$

Table 4: Variable-anisotropic problem on the perturbed-Delaunay mesh. The $p - 3$ rule retains substantially more of the parent-space accuracy on irregular elements.

p	full DOFs	$p - 2$ DOFs	$p - 3$ DOFs	full rel. L^2	$p - 2$ rel. L^2	$p - 3$ rel. L^2
3	359	151	307	3.05×10^{-4}	1.17×10^{-3}	3.29×10^{-4}
4	769	249	561	1.78×10^{-5}	8.97×10^{-5}	2.02×10^{-5}
5	1411	371	891	9.13×10^{-7}	9.13×10^{-6}	1.09×10^{-6}

Table 5: Representative dimension reduction for the two residual rules. The observed quasi-Trefftz dimension is $N_h - \text{rank } \mathbf{R}_h$ with the rank taken from the rank-revealing factorization of Section 7 at the tolerance $\sigma_{\max} \max(\text{shape}) \varepsilon_{\text{mach}}$. In all reported cases $\mathbf{R}_h$ had full row rank, so the observed dimension agrees with the prediction (31).

mesh	p	full DOFs	$p - 2$ DOFs	$p - 3$ DOFs	$p - 2$ frac.	$p - 3$ frac.	check
structured ($n=1$)	3	64	40	58	0.625	0.906	exact
structured ($n=1$)	4	125	65	101	0.520	0.808	exact
structured ($n=1$)	5	216	96	156	0.444	0.722	exact
pert. Delaunay ($n=2$)	3	359	151	307	0.421	0.855	exact
pert. Delaunay ($n=2$)	4	769	249	561	0.324	0.730	exact
pert. Delaunay ($n=2$)	5	1411	371	891	0.263	0.631	exact

so the $m = 2$ rule takes the local complexity from $O(p^2)$ to trace order in the degree.

On structured meshes at $p = 4$, refinement from 2 to 50 triangles reduces the full C^0 relative L^2 error from 1.80×10^{-4} to 7.19×10^{-8} . The $m = 3$ space tracks it throughout, from 2.40×10^{-4} to 9.13×10^{-8} , a ratio between 1.27 and 1.33 at every level, while retaining between 76% and 66% of the parent coordinates. The $m = 2$ space retains between 52% and 32% but its error ratio grows with refinement, from 1.27 to 5.17.

Table 6 gives the degree sweep on 18 triangles. The pattern of the three-dimensional study repeats and sharpens: the $m = 3$ error ratio stays between 1.2 and 1.8 over $p = 3, \dots, 7$, whereas the $m = 2$ ratio grows from 2.1 to 28.1. The measured quasi-Trefftz dimensions agree exactly with the full-row-rank prediction (31) in every row, and the assembled residual map has full row rank in every case computed for this paper.

Two dimensions also allow the effect of element shape to be measured as a sweep rather than at a single mesh. Perturbed-Delaunay meshes of 32 triangles at increasing jitter give a sequence of shape qualities at fixed element count, where quality is the smallest ratio of inradius to circumradius scaled to one for the equilateral triangle. Table 7 reports the resulting error ratios. As the elements degrade from quality 0.79 to 0.66, the $m = 2$ ratio grows monotonically at every degree, by a factor of about two at $p = 6$, while the $m = 3$ ratio moves from 1.74 to 1.82. This is the two-dimensional counterpart of the behavior seen on the perturbed-Delaunay tetrahedral mesh, obtained here as a controlled trend.

Table 6: Two-dimensional variable-anisotropic problem, degree sweep on 18 triangles. The retained dimensions agree exactly with the full-row-rank prediction.

p	full DOFs	retained DOFs		relative L^2			ratio to full	
		$p-2$	$p-3$	full	$p-2$	$p-3$	$p-2$	$p-3$
3	100	46	82	2.79×10^{-5}	5.92×10^{-5}	3.41×10^{-5}	2.12	1.22
4	169	61	115	8.99×10^{-7}	2.50×10^{-6}	1.15×10^{-6}	2.79	1.28
5	256	76	148	2.26×10^{-8}	3.19×10^{-7}	3.61×10^{-8}	14.1	1.60
6	361	91	181	5.05×10^{-10}	3.64×10^{-9}	8.53×10^{-10}	7.20	1.69
7	484	106	214	9.76×10^{-12}	2.74×10^{-10}	1.74×10^{-11}	28.1	1.78

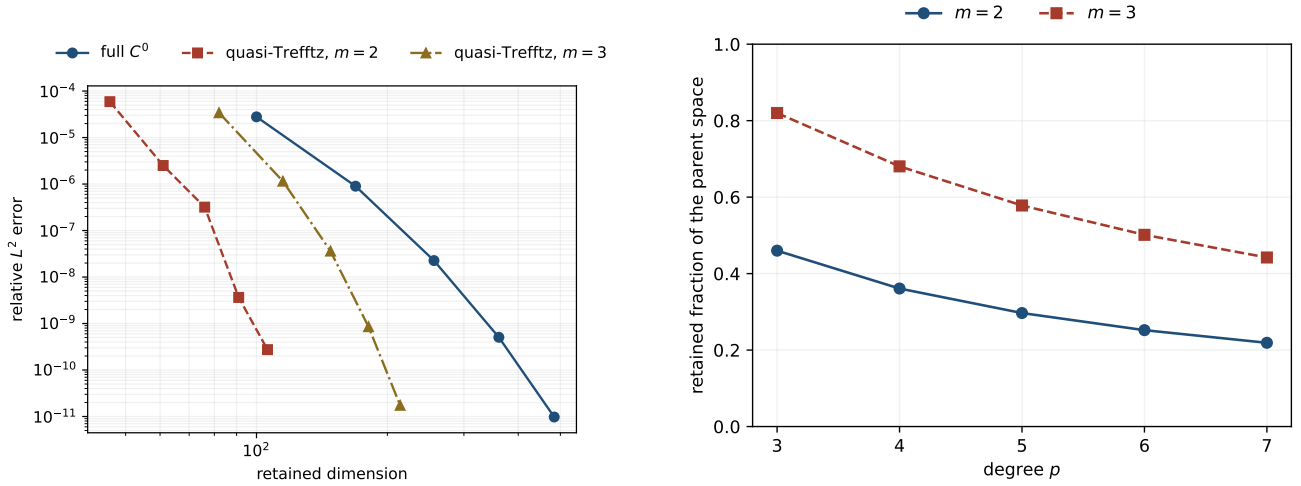


Figure 2: The two-dimensional experiment of Section 8.3, for the operator (1) with the coefficients (38), exact solution $u = \exp(x + y)$, Robin conormal data, on a structured mesh of 18 triangles at degrees $p = 3, 4, 5, 6, 7$. Left: relative L^2 error against retained dimension for the parent space $S_p^0(\Delta)$ and for the reduced spaces with residual offsets $m = 2$ and $m = 3$, the same runs tabulated in Table 6. Right: the ratio of (31) to (7) against degree, which falls from 0.46 to 0.22 for $m = 2$ and from 0.82 to 0.44 for $m = 3$.

Table 7: Ratio of the quasi-Trefftz relative L^2 error to the full C^0 error on 32-triangle perturbed-Delaunay meshes of decreasing shape quality.

quality	$p = 4$		$p = 5$		$p = 6$	
	$p-2$	$p-3$	$p-2$	$p-3$	$p-2$	$p-3$
0.794	8.47	1.32	10.92	1.62	13.05	1.74
0.768	9.41	1.35	10.72	1.64	16.52	1.76
0.732	10.77	1.39	12.21	1.67	18.30	1.78
0.692	12.78	1.43	15.66	1.71	20.96	1.80
0.663	13.79	1.46	18.00	1.74	26.42	1.82

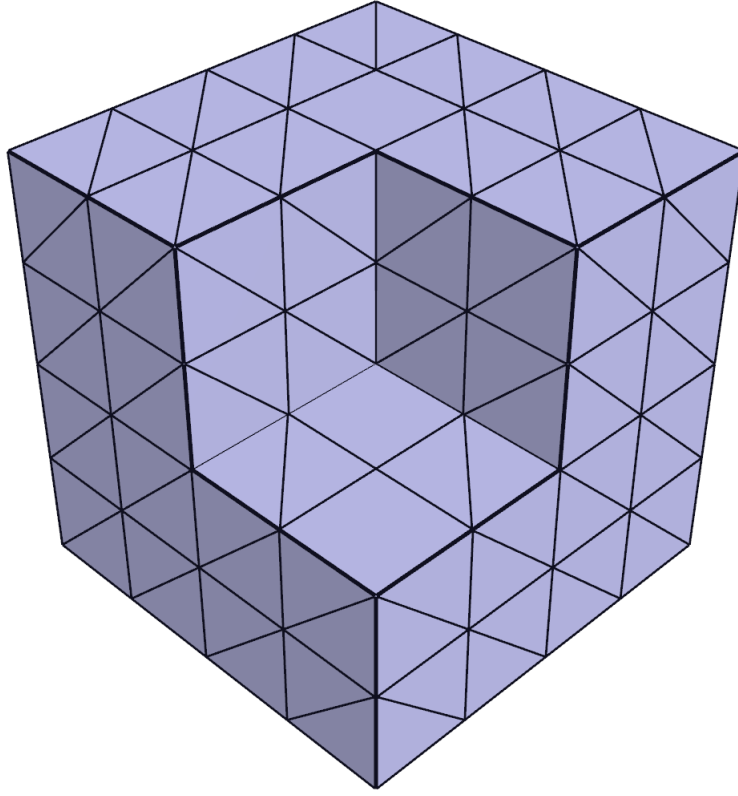


Figure 3: The mesh used in the three-dimensional experiment of Section 8.4. The unit cube is divided $4 \times 4 \times 4$, the octant $[\frac{1}{2}, 1]^3$ is removed, and each of the 56 remaining subcubes is split into six tetrahedra, giving 336 tetrahedra with 117 vertices, 548 edges and 768 faces and hence $\dim S_4^0(\Delta) = 4401$ by (7). The boundary triangulation is drawn, so the diagonals are those of the six-tetrahedron split. The camera looks down the diagonal of the removed octant, so the re-entrant corner at $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ faces the viewer.

8.4. Fichera-corner geometry

Finally we test the construction on a nonconvex three-dimensional geometry. The unit cube is divided $4 \times 4 \times 4$, the octant $[\frac{1}{2}, 1]^3$ is removed, and each of the 56 remaining subcubes is split into six tetrahedra, giving 336 tetrahedra with 117 vertices, 548 edges and 768 faces. We use $p = 4$ and $r = p - 3 = 1$ for the variable anisotropic diffusion–advection–reaction operator (1) with

$$A = \begin{pmatrix} 2 + 0.30x & 0.10 + 0.02z & 0.05 + 0.01y \\ 0.10 + 0.02z & 1.5 + 0.20y & 0.08 + 0.015x \\ 0.05 + 0.01y & 0.08 + 0.015x & 1 + 0.10z \end{pmatrix}, \quad (39)$$

$$b = (0.40 + 0.10y, 0.30 + 0.05z, 0.20 + 0.08x)^T, \quad c = 1 + 0.15xy + 0.05z,$$

and manufacture the smooth oscillatory solution

$$u(x, y, z) = e^{0.15(x+y+z)} \left[\sin(\pi(x + 0.65y - 0.40z) + 0.15) + 0.45 \cos(\pi(1.70x - 0.80y + 1.20z) - 0.25) \right]. \quad (40)$$

Table 8 reports a degree sweep against the parent C^0 Galerkin solution on the same mesh and degree. The assembled dimensions match (7) and the retained dimensions match the full-row-rank formula (31) in every row, and the residual constraints are satisfied to between 10^{-15} and 10^{-12}

Table 8: Fichera-corner degree sweep on 336 tetrahedra. Retained dimensions agree with (31) in every row. The last two columns are the ratios of the quasi-Trefftz errors to those of the parent C^0 Galerkin solution on the same mesh and degree.

p	m	N_h	constraints	retained	qT rel. L^2	parent rel. L^2	L^2 ratio	H^1 ratio
3	2	1981	1344	637	6.45×10^{-2}	9.31×10^{-4}	69.3	59.5
3	3	1981	336	1645	1.02×10^{-3}	9.31×10^{-4}	1.10	1.07
4	2	4401	3360	1041	1.92×10^{-2}	8.10×10^{-5}	236.6	186.2
4	3	4401	1344	3057	1.01×10^{-4}	8.10×10^{-5}	1.25	1.18
5	2	8261	6720	1541	3.20×10^{-3}	6.57×10^{-6}	487.5	375.5
5	3	8261	3360	4901	9.08×10^{-6}	6.57×10^{-6}	1.38	1.28

relative. At $p = 4$ and $m = 3$ the reduction to 69% of the coordinates costs a factor 1.25 in L^2 and 1.18 in H^1 , and the ratio grows only slowly with degree, from 1.10 at $p = 3$ to 1.38 at $p = 5$. The $m = 2$ rule fails badly on this geometry, with ratios of 69, 237 and 487, which is the strongest form of the overcompression seen in Table 4 and explained by the range-lifting measurement of Section 8.5.

8.5. Rank, range stability and quadrature

This subsection tests the three hypotheses the construction rests on.

The deep block.. Proposition 3.2 predicts that $R_{T,I}$ is nonsingular with an h -independent conditioning once the element is small enough. Table 9 confirms this on the planar problem: the condition number of $R_{T,I}$ is constant in h to three digits over a factor 16 of refinement. It is not uniform in the degree, growing by about one order per two degrees, and it depends on the facet by roughly a factor 20 at a fixed degree, which is the spread anticipated in Remark 3.4. At every degree and on every mesh the $m = 2$ block is seven to ten times worse conditioned than the $m = 3$ block, and on perturbed-Delaunay meshes the worst case over all elements and facets at $p = 5$ grows from 2.7×10^3 to 1.0×10^4 for $m = 2$ as the shape quality falls from 0.79 to 0.66, against 3.1×10^2 to 8.3×10^2 for $m = 3$.

Stable range lifting.. Assumption 5.1 asks for a lifting bounded by Ch^2 . Measuring the assembled residual map in the L^2 – L^2 pair, with coefficients through the mass matrix of $S_p^0(\Delta)$ and residual rows through the elementwise Gram matrix of $\mathbb{P}_{p-m}$, the lifting norm is the reciprocal of the smallest nonzero singular value σ_h^+ , so the assumption is the statement that $h^2\sigma_h^+$ stays bounded away from zero. Table 10 reports it. For $m = 3$ the quantity settles to a positive constant, between 102 and 157 over the whole sweep and still decreasing only slowly, so the assumption holds in these experiments. For $m = 2$ it falls steadily under refinement, from 6.97 to 0.62 across the same five meshes, so the assumption fails. This is a quantitative explanation of the accuracy behavior in Tables 4 and 6: the $m = 2$ reduction removes more coordinates than the residual map can stably lift back.

Quadrature.. With the polynomial coefficients used above the residual moments are integrated exactly, so the rank cannot move with the quadrature order. To test the perturbation argument of Section 7 we repeat the measurement with a nonpolynomial coefficient field, taking A and c built from sin and cos of πx , πy and πxy . All element integrals use a Duffy-transformed tensor Gauss–Legendre rule [7], and the order quoted is the number of points per direction. Sweeping

Table 9: Top: condition number of the local deep block $R_{T,I}$ on the planar problem for the first element of a structured mesh, by choice of the facet e . Bottom: condition number of the frozen principal block (15) on the reference triangle, worst case over rotations, against the anisotropy ratio of A . The latter is the quantity entering the threshold (17).

h	p	m	facet 1	facet 2	facet 3
0.707	5	2	5.7×10^2	8.3×10^1	1.5×10^3
0.044	5	2	5.8×10^2	8.2×10^1	1.4×10^3
0.707	5	3	8.2×10^1	3.2×10^1	1.7×10^2
0.044	5	3	8.3×10^1	3.2×10^1	1.7×10^2
0.707	7	3	5.8×10^3	6.7×10^2	1.2×10^4
0.707	8	3	3.6×10^4	3.0×10^3	2.0×10^5
frozen principal block $\kappa(P_T)$, anisotropy ratio 1, 10, 100					
	4	2	1.9×10^1	3.4×10^2	2.9×10^4
	4	3	3.5×10^0	1.6×10^1	3.8×10^1
	6	3	1.2×10^2	2.7×10^3	2.4×10^5
	8	3	2.4×10^3	3.8×10^5	3.2×10^8

that order from p to $2p + 7$ at $p = 5$ and $p = 6$ on 18 triangles, the numerical rank is full at every order and σ_h^+ moves only in the sixth significant digit, from 974.896 at order 5 to 974.898 from order 7 onward. Numerical rank is determined at the same tolerance $\sigma_{\max} \max(\text{shape}) \varepsilon_{\text{mach}}$ used elsewhere, about 2×10^{-13} for these matrices. The rank and the dimension accounting are therefore insensitive to the quadrature over the whole practical range.

8.6. Cost of the constrained solve

The reduction is a statement about representation size, and it should not be confused with a statement about solve cost. The implementation used here solves the constrained problem as a saddle point system with a multiplier for each residual row, which has $N_h + \text{rank } \mathbf{R}_h$ unknowns, more than the parent problem rather than fewer. Table 11 records the comparison on structured planar meshes at $m = 3$.

Three things follow. The saddle system is 1.33 to 1.54 times the parent in unknowns and between three and eleven times the parent in direct solve time, and the ratio worsens with degree. The quasi-Trefftz representation is between 0.46 and 0.65 of the parent, so a method that works in that representation would be smaller, but realizing the saving requires a sparse basis of $\ker \mathbf{R}_h$, which the present prototype does not form; a dense null basis is not an option at these sizes. Finally, the $m = 3$ retained dimension coincides exactly with the parent dimension after static condensation of the element interiors, in every row of Table 11. That is not a coincidence: in two dimensions the element-interior term in (31) cancels, which is (33). The spaces are different, since condensation preserves the parent Galerkin solution while the moment constraints change it, but the coordinate counts agree, and any cost claim for the planar case has to be made against condensation rather than against the raw parent.

8.7. Comparison with a discontinuous parent space

The local count (19) is shared with the quasi-Trefftz discontinuous Galerkin constructions of [13–15] and with embedded Trefftz [20]: all of them remove $\binom{p-m+d}{d}$ directions per element. What differs is the parent space on which that removal acts. A broken parent carries $|\Delta_d| \binom{p+d}{d}$

Table 10: Smallest nonzero singular value of the assembled residual map on structured planar meshes. The quantity is $\sigma_h^+ = \sigma_{\min}^+(M_Y^{-1/2} \mathbf{R}_h M_S^{-1/2})$, where M_S is the mass matrix of $S_p^0(\Delta)$, so that the coefficient norm is $\|\cdot\|_{0,\Omega}$, and M_Y is the block-diagonal elementwise Gram matrix of the Bernstein basis of $\mathbb{P}_{p-m}(T)$, so that the residual norm is the $L^2(\Omega)$ norm of the elementwise projection of the residual. In these norms the lifting of Assumption 5.1 has norm $1/\sigma_h^+$, so the assumption is the statement that the last column is bounded away from zero. Numerical rank is taken at the tolerance $\sigma_{\max} \max(\text{shape})\varepsilon_{\text{mach}}$; the map had full row rank in every row.

triangles	h	p	m	rank $\mathbf{R}_h$	σ_h^+	$h^2\sigma_h^+$
8	0.500	5	3	48	549	137
18	0.333	5	3	108	1079	120
32	0.250	5	3	192	1806	113
50	0.200	5	3	300	2734	109
72	0.167	5	3	432	3862	107
8	0.500	5	2	80	27.9	6.97
18	0.333	5	2	180	24.1	2.68
32	0.250	5	2	320	20.2	1.26
50	0.200	5	2	500	20.5	0.82
72	0.167	5	2	720	22.4	0.62

Table 11: Representation size against solve cost on structured planar meshes at $m = 3$. The saddle point system carries more unknowns than the parent problem, while the quasi-Trefftz representation carries fewer.

triangles	p	parent N_h	constraints	saddle size	qT size	parent (s)	saddle (s)
128	4	1089	384	1473	705	0.0025	0.0066
128	6	2401	1280	3681	1121	0.0079	0.0381
512	4	4225	1536	5761	2689	0.0157	0.0556
512	5	6561	3072	9633	3489	0.0257	0.1415
512	6	9409	5120	14529	4289	0.0493	0.5428

coordinates and retains $|\Delta_d|N_{d,p,m}$ of them, while the assembled parent carries the smaller count (7) and retains (31). Continuity has already removed the duplication of shared coefficients before the equation is imposed, so the two reductions compound.

Table 12 makes the comparison on the meshes used above, with $m = 3$ throughout. In two dimensions the conforming reduced space is between 51% and 57% of the corresponding broken reduced space, and in three dimensions between 31% and 37%. The gap widens with dimension, which is expected: the fraction of coordinates that a simplicial mesh shares between elements grows with d . These counts concern representation size only. A discontinuous formulation buys flexibility that a conforming one does not have, and the comparison is not an accuracy statement.

9. Conclusions

A Bernstein–Bézier representation makes a conformity-first quasi-Trefftz reduction unusually explicit. Relative to the quasi-Trefftz and embedded Trefftz constructions of [13, 20], which reduce a broken space and reconnect it with discontinuous Galerkin terms, the reduction here acts on an already assembled space, so shared coefficients are reduced once and no face parameters enter. The parent space is the ordinary globally assembled C^0 polynomial spline space, and the PDE enters only through residual moments applied after continuity has been enforced. In arbitrary dimension,

Table 12: Retained coordinates for $m = 3$ on an assembled C^0 parent space and on a broken parent space of the same degree and mesh. The last column is the ratio of the two reduced dimensions.

d	mesh	p	C^0 parent	C^0 reduced	broken reduced	ratio
2	18 triangles	3	100	82	162	0.51
2		5	256	148	270	0.55
2		7	484	214	378	0.57
3	52 tetrahedra	3	359	307	988	0.31
3		4	769	561	1612	0.35
3		5	1411	891	2392	0.37

a residual offset m removes $\binom{p-m+d}{d}$ local directions and leaves $\binom{p+d}{d} - \binom{p-m+d}{d}$ free directions. For fixed m this changes the local growth from $O(p^d)$ to $O(p^{d-1})$. The same construction gives exact global reduction formulas in terms of the mesh entity counts and the rank of the assembled residual map.

Ellipticity gives a sufficient condition for the local elimination: the deep block is a perturbation of a positive definite weighted mass matrix whose weight is the conormal quantity $\nabla \lambda_0 \cdot A \nabla \lambda_0$, so it is nonsingular on every facet once the element is small enough. The experiments indicate that $m = 3$ is a useful robust choice, and the range-lifting measurement explains why: the smallest nonzero singular value of the assembled residual map, scaled as Assumption 5.1 requires, is bounded away from zero for $m = 3$ and decays under refinement for $m = 2$. They also show what the reduction does and does not buy: the retained dimension is between 46% and 65% of the parent space, while the saddle point system actually solved is larger than the parent, so realizing the saving needs a sparse null-space basis that the present prototype does not form. On structured meshes a more aggressive $p - 2$ reduction can be effective, but on irregular tetrahedra the $p - 3$ space more reliably follows the accuracy of the parent C^0 method while retaining a substantial reduction in dimension. The Fichera-corner example confirms that the same construction carries directly to nonconvex three-dimensional geometries and a full variable anisotropic second-order operator.

The planar companion problem reproduces the same behavior on triangles and adds a controlled element-quality sweep, in which the $m = 2$ error ratio grows monotonically as shape quality falls while $m = 3$ stays within a factor of two of the parent space. Several extensions are natural. The range-stability condition in Assumption 5.1 can be studied on specific mesh families through C^0 macroelement arguments, while adaptive choices of the retained layers can use local residual rank and approximation information. The same conformity-first principle also extends to higher regularity spaces, where Bernstein–Bézier smoothness relations are imposed before the equation-dependent reduction, as in the C^1 Helmholtz construction of [16].

Data availability

The numerical data underlying the tables and figures are included with the manuscript source package. The reference two- and three-dimensional implementations used for the variable-anisotropic experiments are supplied as supplementary material. A permanent public repository can be assigned at submission or acceptance.

Competing interests

The author declares no competing interests.

AI-assisted manuscript preparation

A generative AI tool was used to assist with organization of the manuscript draft, language editing, and LaTeX preparation. The author is responsible for the mathematical statements, numerical results, source verification, and final manuscript text.

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